

Which Counterexamples to the Petersen Coloring Conjecture Are Colorable Only by Themselves?

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CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

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1 Introduction

1.1 Colorings by a cubic graph

Graphs are finite; they may have parallel edges but no loops. For a vertex v of a graph G , $\partial_G(v)$ is the set of edges at v . Let G and H be cubic graphs. An H -coloring of G is a map $f : E(G) \rightarrow E(H)$ such that for every vertex v of G there is a vertex u of H with $f(\partial_G(v)) = \partial_H(u)$; since both stars have three edges, f is injective on every star. If such a map exists we say that H colors G . The relation is transitive. A *Petersen coloring* is a P -coloring, where P is the Petersen graph, and a cubic graph is 3-edge-colorable if and only if it is colored by the graph on two vertices with three parallel edges.

Jaeger [1] conjectured that every bridgeless cubic graph has a Petersen coloring. The conjecture implies the Berge–Fulkerson conjecture and the 5-cycle double cover conjecture. It was refuted in August 2026: Putman [2] gave two counterexamples G_{112} and H_{112} on 112 vertices with computer-checked certificates, Jooker gave a proof that can be checked by hand, and Goedgebeur, Jooker, Máčajová, Mattiolo, Mazzuoccolo and Ulyanov [3] gave two counterexamples G_{52} and G'_{52} on 52 vertices with a purely theoretical proof, cyclically 4-edge-connected counterexamples of every even order at least 60, and the bounds $40 \leq n_{\min} \leq 52$ for the order $n_{\min}$ of a smallest counterexample. A counterexample G_{68} on 68 vertices is also available from the House of Graphs [7].

1.2 The set $\mathcal{H}_3$

Ma, Mattiolo, Steffen and Wolf [4] proved that there is a unique inclusion-wise minimal set $\mathcal{H}_3$ of connected bridgeless cubic graphs such that every connected bridgeless cubic graph is colored by a member of $\mathcal{H}_3$, and they characterized its members.

Theorem 1.1 ([4], Theorem 3.7, case $r = 3$). *For a connected bridgeless cubic graph G the following are equivalent: $G \in \mathcal{H}_3$; the only connected bridgeless cubic graph that colors G is G itself; no bridgeless cubic graph of smaller order colors G .*

The Petersen coloring conjecture was the statement $\mathcal{H}_3 = \{P\}$. By [4], $\mathcal{H}_3$ is either $\{P\}$ or infinite, so it is now known to be infinite, and its members other than P are all counterexamples: if H colors a counterexample G then H is a counterexample, because a Petersen coloring of H composed with the H -coloring of G would be a Petersen coloring of G . In particular a smallest counterexample belongs to $\mathcal{H}_3$. Goedgebeur et al. ask [3, Section 5.4] whether their counterexamples of order 52 are colorable only by themselves, and observe that the question is tied to whether these graphs are smallest counterexamples.

The question has two informative answers. A bridgeless cubic graph of order less than 52 that colors G_{52} or G'_{52} would be a counterexample below the known upper bound. If no such graph exists, the 52-vertex graph is a member of $\mathcal{H}_3$.

1.3 Results

We first prove two lemmas on the fibers of the vertex map of an H -coloring (Section 2). All fibers have the same parity (Lemma 2.1), and target vertices that are not used by the coloring can be reduced to at most one by the splitting lemma (Lemma 2.3). Together they reduce the question “is G colored by some bridgeless cubic graph of smaller order” to finitely many propositional formulas, one per target order, whose unknown target graph has no part left unconstrained by G (Corollary 2.4). Section 3 describes the formulas and how each answer is certified. Section 4 reports the computations.

2 Fibers of an H -coloring

Throughout this section G is a connected cubic graph of order n , H is a connected cubic graph, f is an H -coloring of G , and $\varphi : V(G) \rightarrow V(H)$ is a *vertex map* of f : a map with $f(\partial_G(v)) = \partial_H(\varphi(v))$ for every v . A vertex map exists by definition, and it is unique unless H is the graph with two vertices and three parallel edges. For $x \in V(H)$ the *fiber* of x is $\varphi^{-1}(x)$ and $n_x = |\varphi^{-1}(x)|$; the vertex x is *used* if $n_x > 0$. We write U for the set of used vertices and $W = V(H) \setminus U$.

Lemma 2.1 (fiber parity). *For every edge $e = xy$ of H , the set $f^{-1}(e)$ is a perfect matching of the subgraph of G induced on $\varphi^{-1}(x) \cup \varphi^{-1}(y)$. Consequently $n_x + n_y$ is even for every edge xy of H , and all the numbers n_x , $x \in V(H)$, have the same parity.*

Proof. A vertex v of G is incident with an edge of $f^{-1}(e)$ if and only if $e \in f(\partial_G(v)) = \partial_H(\varphi(v))$, that is, if and only if $\varphi(v) \in \{x, y\}$. In that case it is incident with exactly one such edge, because f is injective on $\partial_G(v)$. Hence $f^{-1}(e)$ is a matching that covers exactly $\varphi^{-1}(x) \cup \varphi^{-1}(y)$, a set of $n_x + n_y$ vertices. The last statement follows because H is connected. $\square$

So an H -coloring is of one of two kinds. If the fibers are odd, every vertex of H is used and $|V(H)| \leq n$. If the fibers are even, at most $n/2$ vertices of H are used. Both kinds occur: an isomorphism has fibers of size 1, and a 3-edge-coloring of a graph of order $n \equiv 0 \pmod{4}$ has a vertex map with two fibers of size $n/2$.

The second lemma concerns the vertices of H that are not used. We use the splitting lemma of Fleischner [5] in the following form. Let v be a vertex of a graph K and $e_1 = vv_1$, $e_2 = vv_2$ two edges at v . *Splitting e_1 and e_2 off v* means adding a new vertex v^* and replacing e_1, e_2 by edges v^*v_1, v^*v_2 .

Lemma 2.2 ([6], Lemma 1, a consequence of [5]). *Let v be a vertex of degree at least 4 in a bridgeless graph K . There exist edges e_1, e_2 incident with v such that the graph obtained by splitting e_1 and e_2 off v is bridgeless.*

Lemma 2.3 (unused vertices). *Let H be bridgeless and suppose $W \neq \emptyset$. Let m be the number of edges between U and W . Then there is a connected bridgeless cubic graph H' with vertex set $U \cup W'$, where $W' = \emptyset$ if m is even and $|W'| = 1$ if m is odd, and an H' -coloring of G with the same vertex map φ . In particular $|V(H')| \leq |V(H)|$.*

Proof. Every edge of H with an end in U lies in the image of f , since the star of a used vertex is the image of a star of G ; no edge with both ends in W does. Identify W to a single vertex w^* and delete the loops that arise. Every edge cut of the resulting graph H_1 is an edge cut of H , so H_1 is bridgeless; w^* has degree m , and $m \geq 2$ because H is connected and bridgeless. Every other vertex of H_1 is a used vertex, of degree 3, with its star unchanged.

While w^* has degree at least 4, apply Lemma 2.2 to split two edges aw^* , bw^* off w^* so that the graph stays bridgeless. Then $a \neq b$: otherwise the new vertex would be joined to the cubic vertex a by two parallel edges, and the third edge at a would be a bridge. Suppress the new vertex, that is, replace its two edges by a single edge ab . This keeps the graph loopless and bridgeless and lowers the degree of w^* by 2. When the degree of w^* has dropped to 3, stop. When it has dropped to 2, suppress w^* in the same way; its two neighbours are distinct by the same argument. Let H' be the final graph. It is cubic, loopless and bridgeless.

Let μ map every edge of H that has an end in U to the edge of H' that contains it. Two edges merged in one step have their ends other than w^* at distinct vertices, so μ is injective on $\partial_H(u)$ and $\partial_{H'}(u) = \mu(\partial_H(u))$ for every $u \in U$. Hence $\mu \circ f$ is an H' -coloring of G with vertex map φ .

Finally H' is connected. If uv is an edge of G with $\varphi(u) \neq \varphi(v)$, then $f(uv)$ joins $\varphi(u)$ to $\varphi(v)$; it has both ends in U and is therefore an edge of H' . As G is connected, U induces a connected subgraph of H' , and the vertex w^* , if it remains, is joined to U . $\square$

Corollary 2.4. *A connected cubic graph G of order n is colored by a connected bridgeless cubic graph of order less than n if and only if, for some even $k < n$, it has an H -coloring by a connected bridgeless cubic graph H of order k with a vertex map of one of the following kinds:*

- (O) *every vertex of H is used and every fiber is odd;*
- (E0) *every vertex of H is used and every fiber is even, so that $k \leq n/2$;*
- (E1) *exactly one vertex of H is not used and every other fiber is even and nonempty, so that $k - 1 \leq n/2$.*

Proof. Let H of order less than n color G . If every vertex of H is used, Lemma 2.1 gives (O) or (E0). Otherwise the fibers are even by Lemma 2.1, since an unused vertex has an empty fiber, and Lemma 2.3 gives a graph H' of order at most $|V(H)|$ with the same vertex map and at most one unused vertex, which is (E0) or (E1). $\square$

The kinds (O), (E0), (E1) leave no part of H unconstrained by G : every edge of H is the image of an edge of G . For target orders close to n only kind (O) is possible, and the fibers are then almost all singletons. The extreme case of kind (O) can be settled by hand.

Proposition 2.5. *Let G be a connected cubic graph of order $n \geq 8$ without triangles. Then G has no H -coloring by a connected cubic graph H of order $n - 2$ in which every vertex of H is used.*

Proof. Since $n - 2 > n/2$ and every vertex of H is used, the fibers are odd by Lemma 2.1; hence one fiber $\{u_1, u_2, u_3\} = \varphi^{-1}(x)$ has size 3 and all others have size 1. For each of the three edges $e = xy$ at x , the set $f^{-1}(e)$ is a perfect matching on $\{u_1, u_2, u_3\}$ together with the single vertex of the fiber of y (Lemma 2.1), so it contains an edge of G joining two of u_1, u_2, u_3 . The three sets $f^{-1}(e)$ are pairwise disjoint and G is simple, so $u_1 u_2 u_3$ is a triangle of G . $\square$

3 Propositional encoding and certification

For a fixed graph G and a fixed even k we build one formula in conjunctive normal form that is satisfiable if and only if G has an H -coloring of kind (O), (E0) or (E1) by some loopless cubic graph H on k vertices; connectedness and bridgelessness of H are handled separately below. The target graph is part of the unknown.

- *Target.* Every target vertex $i < k$ carries three slots $(i, 0), (i, 1), (i, 2)$. A perfect matching on the $3k$ slots that never pairs two slots of the same vertex is the edge set of a loopless cubic graph on k vertices, and every such graph arises. One variable per admissible slot pair.
- *Map.* Variables $x_{v,i}$ state $\varphi(v) = i$, exactly one i per vertex v . At every v a bijection between the three edges at v and the three slots of $\varphi(v)$ records f on the star of v .
- *Consistency.* For every edge uv of G , the slot assigned to uv at u and the slot assigned to it at v are either the same slot or a matched pair: both describe the edge $f(uv)$.
- *Symmetry breaking.* Target vertices are numbered in the order in which the vertices $0, 1, \dots, n - 1$ of G first use them, and the first vertex of each fiber fixes the order of the three slots.
- *Fibers.* One variable q states that the fibers are odd; for every target vertex an exclusive-or chain forces the parity of its fiber to equal q (Lemma 2.1). The last target vertex but one is used (Lemma 2.3), and q is set to true when $2(k - 1) > n$.

A model is decoded into a pair (H, f) . If H is disconnected or has a bridge, a clause is added that requires a further matched slot pair across the offending vertex partition, which every connected bridgeless target satisfies, and the formula is solved again. The loop ends with a model whose target is connected and bridgeless, or with an unsatisfiable formula.

Certification. A satisfiable answer is checked from the definition by a separate program that reads only G and the decoded pair (H, f) : H is cubic, loopless, connected and bridgeless, and f maps every star of G onto a star of H . An unsatisfiable answer is certified on the final formula (the base formula together with every added clause): it is refuted by CaDiCaL [8] with a DRAT proof, and the proof is checked by `drat-trim` [9]. The incremental solver that drives the loop is not part of the trusted base.

Controls. The formulas with and without the fiber constraints were compared on every even order below the order of the graph for K_4 , the prism, the Petersen graph and the flower snarks J_3 and J_5 (21 instances). The answers agree in every instance. In particular the Petersen graph is colored by no bridgeless cubic graph of smaller order, J_5 is colored by bridgeless cubic graphs of orders 10 and 12 and of no other order below 20, and J_3 is colored by one of order 10 and by none of order 2, 4, 6 or 8.

4 Results

The graphs are G_{52} and G'_{52} (House of Graphs entries 57244 and 57278), both cubic of girth 5 and cyclically 4-edge-connected. Their non-colorability by the Petersen graph was confirmed independently of the published proofs: the edge-map formula with target P is refuted with checked proofs for both graphs, and so is the formula for normal 5-edge-colorings.

In kind (O) with k close to 52 almost all fibers are singletons and the refutation is short; the instances become harder as k decreases and the fibers grow, and the certification of the final formula dominates the running time. No instance required a connectivity or bridge clause: every formula was already unsatisfiable without them, so the statement holds for colorings of kinds (O), (E0), (E1) by arbitrary loopless cubic graphs, connected or not.

5 Remarks and questions

Smallest counterexamples. A smallest counterexample to the Petersen coloring conjecture belongs to $\mathcal{H}_3$, and its order lies between 40 and 52 [3]. Membership of G_{52} and G'_{52} in $\mathcal{H}_3$ is necessary for them to be smallest counterexamples; it is not sufficient, since $\mathcal{H}_3$ may contain graphs of different orders that do not color one another.

The other known counterexamples. The same question is open for the counterexample on 68 vertices and for the two on 112 vertices. An affirmative answer to “is it colored by a smaller bridgeless cubic graph” would exhibit a smaller counterexample, possibly one of G_{52} , G'_{52} . The formulas of Section 3 apply without change; the largest has target order 110.

Why the fiber lemmas matter. Without Lemmas 2.1 and 2.3 the unknown target has vertices and edges that no edge of G constrains. For G_{52} the loop of Section 3 then produced, for each of the ten target orders from 32 to 50, about two thousand targets with a bridge in half an hour, each excluded by one clause, without deciding any order. With the lemmas every edge of the target is the image of an edge of G , and every order was decided without a single such clause.

A question. Is there a counterexample to the Petersen coloring conjecture that is colored by a smaller counterexample? By Lemma 2.1 the fibers of such a coloring are all odd or all even, and by Proposition 2.5 a coloring with every target vertex used needs target order at most $n - 4$ when the graph has no triangles.

Data availability

The edge lists, encoders, checkers, runners, results and the hashes of all formulas and proofs are in the repository https://github.com/fsantibanezleal/CAOS_RESEARCH, under `problems/combinatorics/petersen-coloring/`. The computations of this paper are archived in the folder `experiments/EXP-007-colorable-only-by-itself` of that directory (runner `run_inc.py`, encoder `code/pcclib/hcolor.py`); the controls of Section 3 are in its `artifacts/agreement` folder. The solvers are CaDiCaL 1.7.3 and the CaDiCaL 1.9.5 build of PySAT; proofs were checked with `drat-trim`.

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